

Unit 6

Equations of the First Order but not of the First Degree

Definition

The general first order differential equation of degree $n > 1$ is of the form

$$p_0 \left(\frac{dy}{dx} \right)^n + p_1 \left(\frac{dy}{dx} \right)^{n-1} + p_2 \left(\frac{dy}{dx} \right)^{n-2} + \dots + p_{n-1} \frac{dy}{dx} + p_n = 0 \quad \dots \quad (1)$$

Where, the coefficients $p_0, p_1, p_2, \dots, p_n$ are functions of x or y or constants.

$\frac{dy}{dx}$ is denoted, conventionally by p i.e. $\frac{dy}{dx} = p$. So, the equation (1) takes the form

$$p_0 p^n + p_1 p^{n-1} + p_2 p^{n-2} + \dots + p_{n-1} p + p_n = 0 \quad \dots \quad (2)$$

Such an equation can be solved by different methods, some of which are discussed below.

The solutions of (1) can be presented in the following two forms:

i. The Cartesian Form

This form will consist of x, y and an arbitrary constant c . For example, $y^2 = 2cx + c^2$ is a general solution in the Cartesian Form.

ii. The Parametric Form

This form will consist of two equations of the form $x = f_1(p, c)$ and $y = f_2(p, c)$ where c is an arbitrary constant. These two equations together give parametric equations of the general solution.

This situation arises when the elimination of p is not possible while solving certain equations. Also, we note that p is treated as parameter in this case. Some times this form is also not possible. Then we consider the following forms of solutions in parametric form:

$$\begin{aligned} i. & \quad y = f(x, p); \quad F(x, p, c) = 0 \\ ii. & \quad x = g(y, p); \quad G(y, p, c) = 0 \end{aligned}$$

Method I (Equations Solvable For p)

Let

$$p_0 p^n + p_1 p^{n-1} + p_2 p^{n-2} + \dots + p_{n-1} p + p_n = 0 \quad \dots \quad (1)$$

be the given differential equation of first order and degree $n > 1$. Since (1) is solvable for p it can be put in the form

$$[p - f_1(x, y)][p - f_2(x, y)] \dots [p - f_n(x, y)] = 0 \quad \dots \quad (2)$$

Equating each factor of (2) to zero, we get n equations of the first order and first degree. That is,

$$\begin{aligned} p - f_1(x, y) = 0, p - f_2(x, y) = 0, \dots, p - f_n(x, y) = 0 \\ \Rightarrow p = \frac{dy}{dx} = f_1(x, y), p = \frac{dy}{dx} = f_2(x, y), \dots, p = \frac{dy}{dx} = f_n(x, y) \quad \dots \quad (3) \end{aligned}$$

Let the solutions of these n equations be

$$F_1(x, y, c_1) = 0, F_2(x, y, c_2) = 0, \dots, F_n(x, y, c_n) = 0,$$

where $c_1, c_2, \dots, c_n$ are arbitrary constants of integration.

Since all $c_1, c_2, \dots, c_n$ can have any value, the above solutions will remain general if $c_1, c_2, \dots, c_n$ are replaced by a single arbitrary constant c . So, the above n solutions can be written as

$$F_1(x, y, c) = 0, F_2(x, y, c) = 0 \quad \dots, F_n(x, y, c) = 0.$$

These solutions are combined into a single equation to give the combined solution

$$[F_1(x, y, c)] [F_2(x, y, c)] \dots [F_n(x, y, c)] = 0.$$

Note Since the given equation (1) is of first order, its general solution cannot have more than one arbitrary constant.

Examples

1. Solve $p^2 - 7p + 12 = 0$

Solution

The given equation is

$$\begin{aligned} p^2 - 7p + 12 &= 0 \\ \text{or, } p^2 - 4p - 3p + 12 &= 0 \\ \text{or, } p(p - 4) - 3(p - 4) &= 0 \\ \text{or, } (p - 4)(p - 3) &= 0 \\ \therefore \text{either } p - 4 = 0 \quad \dots (1) \\ \text{or, } p - 3 = 0 \quad \dots (2) \end{aligned}$$

From (1)

$$\begin{aligned} p - 4 &= 0 \\ \rightarrow p &= 4 \\ \rightarrow \frac{dy}{dx} &= 4 \\ \rightarrow dy &= 4dx \end{aligned}$$

Integration implies

$$\begin{aligned} y &= 4x + c \\ y - 4x - c &= 0 \quad \dots (A) \end{aligned}$$

From (2)

$$\begin{aligned} p - 3 &= 0 \\ \rightarrow p &= 3 \\ \rightarrow \frac{dy}{dx} &= 3 \\ \rightarrow dy &= 3dx \end{aligned}$$

Integration implies

$$\begin{aligned} y &= 3x + c \\ y - 3x - c &= 0 \quad \dots (B) \end{aligned}$$

Hence, the required combined solution is

$$(y - 4x - c)(y - 3x - c) = 0,$$

where c is an arbitrary constant.

2. Solve $p^2 - 2pcoshx + 1 = 0$

$$\text{Hint: } coshx = \frac{e^x + e^{-x}}{2}$$

3. Solve $x^2p^2 + xyp - 6y^2 = 0$

Solution 3.

The given equatin is

$$x^2p^2 + xyp - 6y^2 = 0$$

This can be factored as

$$\begin{aligned} x^2p^2 + 3xyp - 2xyp - 6y^2 &= 0 \\ \text{or, } xp(xp + 3y) - 2y(xp + 3y) &= 0 \\ \text{or, } (xp + 3y)(xp - 2y) &= 0 \end{aligned}$$

Therefore, either

$$xp + 3y = 0 \quad \dots \quad (1)$$

Or

$$xp - 2y = 0 \quad \dots \quad (2)$$

From (1)

$$\begin{aligned} xp + 3y &= 0 \\ \rightarrow xp &= -3y \\ \rightarrow x \frac{dy}{dx} &= -3y \\ \rightarrow \frac{dy}{y} &= -3 \frac{dx}{x} \end{aligned}$$

Integration implies

$$\begin{aligned} \int \frac{dy}{y} &= -3 \int \frac{dx}{x} \\ \rightarrow \ln y &= -3 \ln x + \ln c \\ \rightarrow \ln y &= -\ln x^3 + \ln c \\ \rightarrow \ln y + \ln x^3 &= \ln c \\ \rightarrow \ln yx^3 &= \ln c \\ \rightarrow yx^3 &= c \\ \rightarrow yx^3 - c &= 0 \quad \dots \quad (A) \end{aligned}$$

From (2)

$$\begin{aligned} xp - 2y &= 0 \\ \rightarrow xp &= 2y \\ \rightarrow x \frac{dy}{dx} &= 2y \\ \rightarrow \frac{dy}{y} &= \frac{2}{x} dx \end{aligned}$$

Integration implies

$$\begin{aligned} \int \frac{dy}{y} &= 2 \int \frac{1}{x} dx \\ \rightarrow \ln y &= 2 \ln x + \ln c \\ \rightarrow \ln y &= \ln x^2 + \ln c \\ \rightarrow \ln y &= \ln cx^2 \\ \rightarrow y &= cx^2 \\ \rightarrow y - cx^2 &= 0 \quad \dots \quad (B) \end{aligned}$$

Combining (A) and (B), the combined solution is

$$(yx^3 - c)(y - cx^2) = 0,$$

where c is an arbitrary constant.

4. Solve $p^2 + \left(x + y - \frac{2y}{x}\right)p + xy + \frac{y^2}{x^2} - y - \frac{y^2}{x} = 0$

Solution

The given equation is

$$\begin{aligned} p^2 + \left(x + y - \frac{2y}{x}\right)p + xy + \frac{y^2}{x^2} - y - \frac{y^2}{x} &= 0 \\ \rightarrow p^2 + \left[\left(x - \frac{y}{x}\right) + \left(y - \frac{y}{x}\right)\right]p + \left\{y\left(x - \frac{y}{x}\right) - \frac{y}{x}\left(x - \frac{y}{x}\right)\right\} &= 0 \\ \rightarrow p^2 + \left(x - \frac{y}{x}\right)p + \left(y - \frac{y}{x}\right)p + \left(x - \frac{y}{x}\right)\left(y - \frac{y}{x}\right) &= 0 \end{aligned}$$

$$\begin{aligned} \rightarrow p \left[p + x - \frac{y}{x} \right] + \left(y - \frac{y}{x} \right) \left[p + x - \frac{y}{x} \right] &= 0 \\ \rightarrow \left[p + x - \frac{y}{x} \right] \left[p + y - \frac{y}{x} \right] &= 0 \end{aligned}$$

Therefore, either

$$p + x - \frac{y}{x} = 0 \quad \dots \quad (1)$$

Or,

$$p + y - \frac{y}{x} = 0 \quad \dots \quad (2)$$

From (1),

$$\begin{aligned} p + x - \frac{y}{x} &= 0 \\ \text{or, } \frac{dy}{dx} + x - \frac{y}{x} &= 0 \\ \text{or, } \frac{dy}{dx} - \frac{y}{x} &= -x \quad \dots \quad (3) \end{aligned}$$

which is linear with $p = -\frac{1}{x}$.

$$\therefore I.F. = e^{\int p dx} = e^{\int -\frac{1}{x} dx} = e^{-\ln x} = e^{\ln \frac{1}{x}} = \frac{1}{x}.$$

$\therefore$ multiplying both sides by $\frac{1}{x}$, (3) reduces to

$$\begin{aligned} \frac{1}{x} \frac{dy}{dx} - \frac{y}{x^2} &= -1 \\ \rightarrow \frac{d}{dx} \left(\frac{y}{x} \right) &= -1 \\ \rightarrow d \left(\frac{y}{x} \right) &= -1 dx \end{aligned}$$

Integrating, we get

$$\begin{aligned} \frac{y}{x} &= -x + C \\ \rightarrow y &= -x^2 + Cx \\ \rightarrow y + x^2 - Cx &= 0 \quad \dots \quad (A) \end{aligned}$$

From (2),

$$\begin{aligned} p + y - \frac{y}{x} &= 0 \\ \rightarrow \frac{dy}{dx} + y - \frac{y}{x} &= 0 \\ \rightarrow \frac{dy}{dx} + y \left(1 - \frac{1}{x} \right) &= 0 \\ \rightarrow \frac{dy}{dx} &= -y \left(1 - \frac{1}{x} \right) \\ \rightarrow \frac{dy}{y} &= \left(\frac{1}{x} - 1 \right) dx \end{aligned}$$

Integration implies

$$\int \frac{dy}{y} = \int \left(\frac{1}{x} - 1 \right) dx$$

$$\begin{aligned}
&\rightarrow \int \frac{dy}{y} = \int \left(\frac{1}{x} - 1 \right) dx \\
&\rightarrow \ln y = \ln x - x + C \\
&\rightarrow \ln y = \ln x - x + \ln C \\
&\rightarrow \ln y = \ln Cx - x \\
&\rightarrow \ln \frac{y}{Cx} = -x \\
&\rightarrow \frac{y}{Cx} = e^{-x} \\
&\rightarrow y = Cxe^{-x} \\
&\rightarrow y - Cxe^{-x} = 0 \quad \dots \quad (B)
\end{aligned}$$

Hence the combined solution of the given equation is

$$(y + x^2 - Cx)(y - Cxe^{-x}) = 0,$$

where C is some arbitrary constant.

5. Solve $x^2p^2 - 2xyp + y^2 - x^2y^2 - x^4 = 0$

Solution

The given equation is

$$\begin{aligned}
&x^2p^2 - 2xyp + y^2 - x^2y^2 - x^4 = 0 \\
&\rightarrow (xp - y)^2 = x^2y^2 + x^4 \\
&\rightarrow (xp - y)^2 = x^2(y^2 + x^2) \\
&\rightarrow xp - y = \pm x\sqrt{y^2 + x^2} \\
&\rightarrow x \frac{dy}{dx} - y = \pm x\sqrt{y^2 + x^2} \\
&\rightarrow xdy - ydx = \pm x\sqrt{y^2 + x^2}dx \\
&\rightarrow \frac{xdy - ydx}{x^2} = \pm \frac{\sqrt{y^2 + x^2}}{x} dx \\
&\rightarrow d\left(\frac{y}{x}\right) = \pm \sqrt{\frac{y^2}{x^2} + 1} dx \\
&\rightarrow d\left(\frac{y}{x}\right) = \pm \sqrt{\left(\frac{y}{x}\right)^2 + 1} dx \\
&\rightarrow \frac{d\left(\frac{y}{x}\right)}{\sqrt{\left(\frac{y}{x}\right)^2 + 1}} = \pm dx
\end{aligned}$$

On LHS, put $\frac{y}{x} = t$, then we have $d\left(\frac{y}{x}\right) = dt$.

Therefore,

$$\frac{dt}{\sqrt{t^2 + 1}} = \pm dx$$

Integrating, we get

$$\begin{aligned}
&\int \frac{dt}{\sqrt{t^2 + 1}} = \pm \int dx \\
&\rightarrow \sin^{-1}t = \pm x + C \\
&\rightarrow \sin^{-1}\left(\frac{y}{x}\right) = \pm x + C
\end{aligned}$$

$$\begin{aligned}\rightarrow \frac{y}{x} &= \sin(C \pm x) \\ \rightarrow y &= x \sin(C \pm x) \\ \rightarrow y - x \sin(C \pm x) &= 0\end{aligned}$$

Hence, the combined solution is

$$[y - x \sin(C + x)][y - x \sin(C - x)] = 0 \quad \blacksquare$$

6. Solve $\left(\frac{dy}{dx}\right)^2 + 2y \cot x \frac{dy}{dx} = y^2$

Solution

The given equation is

$$\begin{aligned}\left(\frac{dy}{dx}\right)^2 + 2y \cot x \frac{dy}{dx} &= y^2 \\ \text{or, } p^2 + 2y \cot x p - y^2 &= 0\end{aligned}$$

which is quadratic in p , so

$$\begin{aligned}p &= \frac{-2y \cot x \pm \sqrt{(2y \cot x)^2 - 4 \cdot 1 \cdot (-y^2)}}{2} \\ \text{or, } p &= \frac{-2y \cot x \pm \sqrt{4y^2 \cot^2 x + 4y^2}}{2} \\ \text{or, } p &= \frac{-2y \cot x \pm 2y \sqrt{\cot^2 x + 1}}{2} \\ \text{or, } p &= \frac{-2y \cot x \pm 2y \operatorname{cosec} x}{2} \\ \text{or, } p &= -y \cot x \pm y \operatorname{cosec} x \\ \text{or, } \frac{dy}{dx} &= y(-\cot x \pm \operatorname{cosec} x) \\ \text{or, } \frac{dy}{y} &= (-\cot x \pm \operatorname{cosec} x) dx\end{aligned}$$

Therefore, either

$$\frac{dy}{y} = (-\cot x + \operatorname{cosec} x) dx \quad \dots \quad (1)$$

Or,

$$\frac{dy}{y} = (-\cot x - \operatorname{cosec} x) dx \quad \dots \quad (2)$$

From (1), integration implies

$$\begin{aligned}\int \frac{dy}{y} &= \int (-\cot x + \operatorname{cosec} x) dx \\ \rightarrow \ln y &= -\int \cot x dx + \int \operatorname{cosec} x dx \\ \rightarrow \ln y &= -\ln \sin x + \log(\operatorname{cosec} x - \cot x) + \ln C \\ \rightarrow \ln y - \ln C &= \log \frac{(\operatorname{cosec} x - \cot x)}{\sin x} \\ \rightarrow \ln \frac{y}{C} &= \ln \left[\left(\frac{1}{\sin x} - \frac{\cos x}{\sin x} \right) \times \frac{1}{\sin x} \right] \\ \rightarrow \ln \frac{y}{C} &= \ln \left[\left(\frac{1 - \cos x}{\sin^2 x} \right) \right] \\ \rightarrow \frac{y}{C} &= \left[\left(\frac{1 - \cos x}{\sin^2 x} \right) \right]\end{aligned}$$

$$\begin{aligned}
&\rightarrow y \sin^2 x = C(1 - \cos x) \\
&\rightarrow y - y \cos^2 x = C(1 - \cos x) \\
&\rightarrow y - y \cos^2 x + C \cos x - C = 0 \quad \dots \quad (A)
\end{aligned}$$

From (2), integration implies

$$\begin{aligned}
&\int \frac{dy}{y} = \int (-\cot x - \operatorname{cosec} x) dx \\
&\rightarrow \ln y = - \int \cot x dx - \int \operatorname{cosec} x dx \\
&\rightarrow \ln y = -\ln \sin x - \ln(\operatorname{cosec} x - \cot x) + \ln C \\
&\rightarrow \ln y - \ln C = -\ln \sin x (\operatorname{cosec} x - \cot x) \\
&\rightarrow \ln y - \ln C = -\ln(1 - \cos x) \\
&\rightarrow \ln \frac{y}{C} = \ln \left[\left(\frac{1}{1 - \cos x} \right) \right] \\
&\rightarrow \frac{y}{C} = \frac{1}{1 - \cos x} \\
&\rightarrow y - y \cos x = C \\
&\rightarrow y - y \cos x - C = 0 \quad \dots \quad (B)
\end{aligned}$$

Hence, the combined solution is

$$[y - y \cos^2 x + C \cos x - C][y - y \cos x - C] = 0,$$

where C is some arbitrary constant.

Method II (Equations Solvable for x)

If the equation

$$\begin{aligned}
&f(x, y, p) = 0, i. e. \\
&p_0 p^n + p_1 p^{n-1} + p_2 p^{n-2} + \dots + p_{n-1} p + p_n = 0 \quad \dots \quad (1)
\end{aligned}$$

is solvable for x, we can express x explicitly in term of y and p. Thus, an equation solvable for x can be reduced to the form

$$x = F(y, p) \quad \dots \quad (2)$$

Differentiating (1) w.r.t. y, we get

$$\begin{aligned}
&\frac{dx}{dy} = \phi \left(y, p, \frac{dp}{dy} \right) \\
&\rightarrow \frac{1}{p} = \phi \left(y, p, \frac{dp}{dy} \right) \quad \dots \quad (3)
\end{aligned}$$

which is an equation with two variables y and p. Let its solution be

$$\psi(y, p, c) = 0 \quad \dots \quad (4)$$

where c is an arbitrary constant.

Eliminating p between (2) and (4), we get the solution of (1) as

$$g(x, y, c) = 0.$$

On the other hand, if p can not be eliminated from (2) and (4), we solve them to express x and y in terms of p and c in the following form

$$x = f_1(p, c); \quad y = f_2(p, c) \quad \dots \quad (5)$$

These two equations, taken together, constitute the general solution of (2) in parametric form, p being a parameter.

Some times, it is not possible to write the solution in the parametric form (5). In such cases (2) and (4) themselves together are said to form the solution in parametric form.

Special Case of Method II

(Equations That do not Contain y)

If the given equation does not involve y , it can be put in the form

$$f(x, p) = 0 \quad \dots \quad (6)$$

If (6) is solvable for p , it can be reduced to the form

$$p = F(x) \quad \text{or} \quad \frac{dy}{dx} = F(x)$$

which is easily integrable to find the required solution.

If (6) is solvable for x , it can be put in the form

$$x = G(p),$$

which is of the form (2) and can be solved as before.

Examples

1. Solve $y = 2px + y^2p^3$

Solution

The given equation is

$$y = 2px + y^2p^3 \quad \dots \quad (1)$$

Solving (1) for x we get

$$x = \frac{y}{2p} - \frac{y^2p^2}{2} \quad \dots \quad (2)$$

Differentiating (2) w.r.t. y , we get

$$\begin{aligned} \frac{dx}{dy} &= \frac{1}{2p} - \frac{ydp}{2p^2dy} - \frac{2yp^2}{2} - \frac{y^2}{2} \cdot 2p \frac{dp}{dy} \\ \text{or,} \quad \frac{1}{p} &= \frac{1}{2p} - \frac{ydp}{2p^2dy} - yp^2 - y^2p \frac{dp}{dy} \\ \text{or,} \quad \frac{1}{2p} &= \frac{dp}{dy} \left(-\frac{y}{2p^2} - y^2p \right) - yp^2 \\ \text{or,} \quad \frac{1}{2p} + yp^2 &= \left(-\frac{y}{2p^2} - y^2p \right) \frac{dp}{dy} \\ \text{or,} \quad p \left(\frac{1}{2p^2} + yp \right) &= -y \left(\frac{1}{2p^2} + yp \right) \frac{dp}{dy} \\ \text{or,} \quad p \left(\frac{1}{2p^2} + yp \right) + y \left(\frac{1}{2p^2} + yp \right) \frac{dp}{dy} &= 0 \\ \text{or,} \quad \left(\frac{1}{2p^2} + yp \right) \left(p + y \frac{dp}{dy} \right) &= 0 \end{aligned}$$

Therefore, either

$$\left(\frac{1}{2p^2} + yp \right) = 0 \quad \dots \quad (3)$$

Or,

$$\left(p + y \frac{dp}{dy} \right) = 0 \quad \dots \quad (4)$$

We neglect the factor which does not involve $\frac{dp}{dy}$. So, we consider the equation (4) only.

$$\begin{aligned} p + y \frac{dp}{dy} &= 0 \\ \text{or,} \quad \frac{1}{p} dp + \frac{1}{y} dy &= 0 \end{aligned}$$

Integration implies

$$\begin{aligned}\int \frac{1}{p} dp + \int \frac{1}{y} dy &= 0 \\ \rightarrow \ln p + \ln y &= \ln C \\ \rightarrow \ln py &= \ln C \\ \rightarrow py &= C \\ \rightarrow p &= \frac{C}{y}\end{aligned}$$

Substituting this value of p in (1) we get the required solution as

$$\begin{aligned}y &= 2\frac{C}{y}x + y^2\left(\frac{C}{y}\right)^3 \\ \rightarrow y^2 &= 2Cx + C^3 \quad \blacksquare\end{aligned}$$

2. Solve $x = y + a \ln p$

Solution

This given equation is

$$x = y + a \ln p \quad \dots \quad (1)$$

Differentiating (1) w.r.t. y , we get

$$\begin{aligned}\frac{dx}{dy} &= 1 + a \cdot \frac{1}{p} \cdot \frac{dp}{dy} \\ \rightarrow \frac{1}{p} &= 1 + \frac{a}{p} \cdot \frac{dp}{dy} \\ \rightarrow 1 &= p + a \cdot \frac{dp}{dy} \\ \rightarrow 1 - p &= a \cdot \frac{dp}{dy} \\ \rightarrow dy &= \frac{a}{1-p} dp\end{aligned}$$

Integration implies

$$\begin{aligned}\int dy &= \int \frac{a}{1-p} dp \\ \rightarrow y &= c - a \ln(1-p) \\ \therefore y &= c - a \ln(1-p) \quad \dots \quad (2)\end{aligned}$$

where c is an arbitrary constant of integration.

Using (2) in (1), we get

$$\begin{aligned}x &= c - a \ln(1-p) + a \ln p \\ x &= c - a \ln\left(\frac{1-p}{p}\right) \quad \dots \quad (3)\end{aligned}$$

Equation (2) and (3) constitute a general solution of (1) in parametric form with p as a parameter.

3. Solve $x + \frac{p}{\sqrt{1+p^2}} = a$

Solution

The given equation is

$$\begin{aligned}x + \frac{p}{\sqrt{1+p^2}} &= a \\ x - a &= -\frac{p}{\sqrt{1+p^2}} \quad \dots \quad (1)\end{aligned}$$

Differentiating (1) w.r.t. y , we get

$$\begin{aligned}\frac{dx}{dy} &= - \left[\frac{\sqrt{1+p^2} \cdot 1 - p \cdot \frac{1}{2\sqrt{1+p^2}} \cdot 2p}{(\sqrt{1+p^2})^2} \right] \frac{dp}{dy} \\ \text{or, } \frac{1}{p} &= - \left[\frac{\sqrt{1+p^2} - \frac{p^2}{\sqrt{1+p^2}}}{1+p^2} \right] \frac{dp}{dy} \\ \text{or, } \frac{1}{p} &= - \left[\frac{\frac{1+p^2-p^2}{\sqrt{1+p^2}}}{1+p^2} \right] \frac{dp}{dy} \\ \text{or, } \frac{1}{p} &= - \left[\frac{1}{\sqrt{1+p^2}} \times \frac{1}{1+p^2} \right] \frac{dp}{dy} \\ \text{or, } \frac{1}{p} &= - \left[\frac{1}{(1+p^2)^{\frac{3}{2}}} \right] \frac{dp}{dy} \\ \text{or, } dy &= - \frac{p}{(1+p^2)^{\frac{3}{2}}} dp\end{aligned}$$

Integrating, we get

$$\begin{aligned}\int dy &= - \int \frac{p}{(1+p^2)^{\frac{3}{2}}} dp \\ \rightarrow y &= - \int \frac{p}{(1+p^2)^{\frac{3}{2}}} dp\end{aligned}$$

Put

$$\begin{aligned}1+p^2 &= t \\ 2pdp &= dt \\ \therefore pdp &= \frac{dt}{2}\end{aligned}$$

$$\begin{aligned}\text{Thus, } y &= - \int \frac{p}{(1+p^2)^{\frac{3}{2}}} dp = - \int \frac{\frac{dt}{2}}{t^{\frac{3}{2}}} \\ \rightarrow y &= - \frac{1}{2} \int t^{-\frac{3}{2}} dt \\ \rightarrow y &= - \frac{1}{2} \frac{t^{-\frac{1}{2}}}{-\frac{1}{2}} + C \\ \rightarrow y &= \frac{1}{\sqrt{t}} + C \\ \rightarrow y &= \frac{1}{\sqrt{1+p^2}} + C \quad \dots \quad (2) \\ \rightarrow y - C &= \frac{1}{\sqrt{1+p^2}}\end{aligned}$$

From (1) and (2), squaring and adding, we get

$$(x-a)^2 + (y-C)^2 = \left(-\frac{p}{\sqrt{1+p^2}}\right)^2 + \left(\frac{1}{\sqrt{1+p^2}}\right)^2$$

$$\rightarrow (x-a)^2 + (y-C)^2 = \frac{p^2}{1+p^2} + \frac{1}{1+p^2} = \frac{p^2+1}{1+p^2} = 1$$

Hence, the required solution in Cartesian form is

$$(x-a)^2 + (y-C)^2 = 1,$$

where C is an arbitrary constant of integration.

4. Solve $p^2 - 2xp + 1 = 0$

Solution

The given equation is

$$p^2 - 2xp + 1 = 0.$$

This can be written as

$$x = \frac{p^2 + 1}{2p} \quad \dots \quad (1)$$

Differentiating w.r.t y , we get

$$\frac{dx}{dy} = \frac{d\left(\frac{p^2 + 1}{2p}\right)}{dy}$$

$$\text{or, } \frac{1}{p} = \frac{d\left(\frac{p^2 + 1}{2p}\right)}{dp} \times \frac{dp}{dy}$$

$$\text{or, } \frac{1}{p} = \frac{2p \frac{d(p^2 + 1)}{dp} - (p^2 + 1) \frac{d(2p)}{dp}}{(2p)^2} \times \frac{dp}{dy}$$

$$\text{or, } \frac{1}{p} = \frac{2p \times 2p - (p^2 + 1) \times 2}{4p^2} \times \frac{dp}{dy}$$

$$\text{or, } \frac{1}{p} = \frac{4p^2 - 2p^2 - 2}{4p^2} \times \frac{dp}{dy}$$

$$\text{or, } \frac{1}{p} = \frac{2p^2 - 2}{4p^2} \times \frac{dp}{dy}$$

$$\text{or, } 1 = \frac{p^2 - 1}{2p} \times \frac{dp}{dy}$$

$$\text{or, } dy = \left(\frac{p^2 - 1}{2p}\right) dp$$

$$\text{or, } dy = \frac{1}{2} \left(p - \frac{1}{p}\right) dp$$

Integration implies

$$y = \frac{1}{2} \int \left(p - \frac{1}{p}\right) dp$$

$$y = \frac{1}{2} \left(\frac{p^2}{2} - \ln p\right) + C$$

$$y = \frac{p^2}{4} - \frac{1}{2} \ln p + C \quad \dots \quad (2)$$

Hence, the equations (1) and (2) constitute parametric solution of the equation with p as a parameter.

Method III [Equation Solvable for y]

If the equation $f(x, y, p) = 0$ is solvable for y , we can express y explicitly in terms of x and p . That is, an equation $f(x, y, p) = 0$, solvable for y , can be written as

$$y = F(x, p) \quad \dots \quad (1)$$

Differentiating (1) w.r.t. x , we get

$$\begin{aligned} \frac{dy}{dx} &= \varphi\left(x, p, \frac{dy}{dx}\right) \\ \text{i.e. } p &= \varphi\left(x, p, \frac{dy}{dx}\right) \quad \dots \quad (2) \end{aligned}$$

which is an equation involving two variables x and p .

Let its solution be

$$\psi(x, p, c) = 0 \quad \dots \quad (3),$$

where c is an arbitrary constant. Eliminating p between (1) and (3), the solution of (1) is obtained in the form

$$g(x, y, c) = 0.$$

If it is not possible to eliminate p between (1) and (3), we solve (1) and (3) to express x and y in terms of p and c in the following forms:

$$x = f_1(p, c); \quad y = f_2(p, c) \quad \dots \quad (4)$$

These two equations together form the general solution of (1) in parametric form with p as parameter. Some times, even the form (4) of the solution is not possible. In that case (1) and (3) themselves are said to form the solution in parametric form.

Two Special Cases

Case I [Lagrange's Equation]

A differential equation of the form

$$y = xF(p) + f(p) \quad \dots \quad (5)$$

is known as Lagrange's equation. To solve it, we proceed as follows:

Differentiate (5) w.r.t. x , we get

$$\begin{aligned} \frac{dy}{dx} &= F(p) + xF'(p) \frac{dp}{dx} + f'(p) \frac{dp}{dx} \\ p &= F(p) + xF'(p) \frac{dp}{dx} + f'(p) \frac{dp}{dx} \\ p - F(p) &= [xF'(p) + f'(p)] \frac{dp}{dx} \\ \frac{dx}{dp} &= \frac{[xF'(p) + f'(p)]}{p - F(p)} \\ \frac{dx}{dp} - \frac{F'(p)}{p - F(p)} x &= \frac{f'(p)}{p - F(p)}, \end{aligned}$$

which is linear in x and can be solved as usual to give a solution of the form

$$x = \phi(p, c) \quad \dots \quad (6)$$

We now eliminate p between (5) and (6) to get the required solution.

If p can not be eliminated, then by putting value of x in (5) we get

$$y = \phi(p, c)F(p) + f(p) \quad \dots \quad (7)$$

Equations (6) and (7) together form the required solution in parametric form with p as a parameter.

Case II [Equation That do not Contain x]

If the given differential equation does not contain x , it can be put in the form

$$f(y, p) = 0 \quad \dots \quad (8)$$

If (8) is solvable for p , it reduces to

$$p = F(y) \quad \text{or} \quad \frac{dy}{dx} = F(y),$$

which can be easily solved to give the required solution.

If (8) is solvable for y , it can be put in the form

$$y = G(p),$$

which is of the form (5) and can be solved as before.

Examples:

1. Solve $y = 3x + \ln p$

Solution

The given equation is

$$y = 3x + \ln p \quad \dots \quad (1)$$

Differentiating w.r.t x , we get

$$\begin{aligned} \frac{dy}{dx} &= 3 + \frac{1}{p} \cdot \frac{dp}{dx} \\ \text{or,} \quad p &= 3 + \frac{1}{p} \cdot \frac{dp}{dx} \\ \text{or,} \quad p - 3 &= \frac{1}{p} \cdot \frac{dp}{dx} \\ \text{or,} \quad dx &= \frac{1}{p(p-3)} dp \end{aligned}$$

Integration implies

$$\begin{aligned} \int dx &= \int \frac{1}{p(p-3)} dp \\ \text{or,} \quad x &= \frac{1}{3} \int \left(\frac{1}{p-3} - \frac{1}{p} \right) dp \\ \text{or,} \quad x &= \frac{1}{3} [\ln(p-3) - \ln p] - \frac{1}{3} \ln C \\ \text{or,} \quad 3x &= \ln \frac{(p-3)}{pC} \\ \text{or,} \quad e^{3x} &= \frac{(p-3)}{pC} \\ \text{or,} \quad pCe^{3x} &= p-3 \\ \text{or,} \quad p - pCe^{3x} &= 3 \\ \text{or,} \quad p(1 - Ce^{3x}) &= 3 \\ \text{or,} \quad p &= \frac{3}{(1 - Ce^{3x})} \end{aligned}$$

Substituting this value of p in (1) we get

$$y = 3x + \ln \frac{3}{(1 - Ce^{3x})},$$

where C is an arbitrary constant.

Therefore, the required general solution is

$$y = 3x + \ln \frac{3}{(1 - Ce^{3x})}.$$

2. Solve $y = yp^2 + 2px$

Solution

The given equation is

$$y = yp^2 + 2px \quad \dots \quad (1)$$

Solving it for y,

$$\begin{aligned} y(1 - p^2) &= 2px \\ y &= \frac{2px}{(1 - p^2)} \quad \dots \quad (2) \end{aligned}$$

Differentiating it w.r.t. x, we get

$$\begin{aligned} \frac{dy}{dx} &= \frac{\left[(1 - p^2) \left\{ 2p + 2x \frac{dp}{dx} \right\} - 2px(-2p) \frac{dp}{dx} \right]}{(1 - p^2)^2} \\ \text{or, } p &= \frac{\left[2p(1 - p^2) + 2x(1 - p^2) \frac{dp}{dx} + 4p^2x \frac{dp}{dx} \right]}{(1 - p^2)^2} \\ \text{or, } p(1 - p^2)^2 &= 2p(1 - p^2) + 2x(1 - p^2) \frac{dp}{dx} + 4p^2x \frac{dp}{dx} \\ \text{or, } p(1 - p^2)\{1 - p^2 - 2\} - 2x\{1 - p^2 + 2p^2\} \frac{dp}{dx} &= 0 \\ \text{or, } p(1 + p^2)(p^2 - 1) - 2x(1 + p^2) \frac{dp}{dx} &= 0 \\ \text{or, } (1 + p^2) \left[p(p^2 - 1) - 2x \frac{dp}{dx} \right] &= 0 \quad \dots \quad (3) \end{aligned}$$

Neglecting the factor which does not contain $\frac{dp}{dx}$, (3) reduces to

$$\begin{aligned} p(p^2 - 1) - 2x \frac{dp}{dx} &= 0 \\ \text{or, } p(p^2 - 1) &= 2x \frac{dp}{dx} \\ \text{or, } \frac{2}{p(p^2 - 1)} dp &= \frac{dx}{x} \\ \text{or, } \left[\frac{1}{p-1} + \frac{1}{p+1} - \frac{2}{p} \right] dp &= \frac{dx}{x} \end{aligned}$$

Integration implies

$$\begin{aligned} \int \left[\frac{1}{p-1} + \frac{1}{p+1} - \frac{2}{p} \right] dp &= \int \frac{dx}{x} \\ \text{or, } \int \frac{1}{p-1} dp + \int \frac{1}{p+1} dp - 2 \int \frac{1}{p} dp &= \int \frac{dx}{x} \\ \text{or, } \ln(p-1) + \ln(p+1) - 2\ln p &= \ln x + \ln C \\ \text{or, } \ln(p-1)(p+1) - \ln p^2 &= \ln Cx \\ \text{or, } \ln \frac{(p-1)(p+1)}{p^2} &= \ln Cx \\ \text{or, } \frac{p^2 - 1}{p^2} &= Cx \\ \text{or, } p^2 - 1 &= p^2 Cx \\ \text{or, } p^2(1 - Cx) &= 1 \\ \text{or, } p^2 &= \frac{1}{1 - Cx} \end{aligned}$$

$$\therefore p = \frac{1}{\sqrt{1-Cx}}$$

Putting this value of p in (1), we get

$$\begin{aligned} y &= \frac{y}{1-Cx} + \frac{2x}{\sqrt{1-Cx}} \\ \text{or, } y \left(1 - \frac{1}{1-Cx}\right) &= \frac{2x}{\sqrt{1-Cx}} \\ \text{or, } y \left(\frac{1-Cx-1}{1-Cx}\right) &= \frac{2x}{\sqrt{1-Cx}} \\ \text{or, } \frac{-Cxy}{1-Cx} &= \frac{2x}{\sqrt{1-Cx}} \\ \text{or, } \frac{-Cy}{\sqrt{1-Cx}} &= 2 \\ \text{or, } -Cy &= 2\sqrt{1-Cx} \\ \text{or, } C^2y^2 &= 4(1-Cx) \end{aligned}$$

Therefore, the required solution of (1) is

$$C^2y^2 = 4(1-Cx),$$

where C is some arbitrary constant.

3. Solve $y = x + \operatorname{atan}^{-1} p$

Solution

The given equation is

$$y = x + \operatorname{atan}^{-1} p \quad \dots \quad (1)$$

Differentiating w.r.t. x , we get

$$\begin{aligned} \frac{dy}{dx} &= 1 + a \cdot \frac{1}{1+p^2} \frac{dp}{dx} \\ \text{or, } p &= 1 + \frac{a}{1+p^2} \frac{dp}{dx} \\ \text{or, } p-1 &= \frac{a}{1+p^2} \frac{dp}{dx} \\ \text{or, } dx &= \frac{a}{(p-1)(1+p^2)} dp \\ \text{or, } dx &= \frac{a}{2} \left[\frac{1}{p-1} - \frac{1+p}{1+p^2} \right] dp \\ \text{or, } dx &= \frac{a}{2} \left[\frac{1}{p-1} - \frac{1}{1+p^2} - \frac{p}{1+p^2} \right] dp \end{aligned}$$

Integration implies

$$\begin{aligned} \int dx &= \frac{a}{2} \int \left[\frac{1}{p-1} - \frac{1}{1+p^2} - \frac{p}{1+p^2} \right] dp \\ x &= \frac{a}{2} \left[\ln(p-1) - \tan^{-1} p - \frac{1}{2} \ln(1+p^2) \right] + C \\ x &= \frac{a}{2} \left[\ln(p-1) - \tan^{-1} p - \ln \sqrt{(1+p^2)} \right] + C \\ x &= \frac{a}{2} \left[\ln \frac{(p-1)}{\sqrt{(1+p^2)}} - \tan^{-1} p \right] + C \quad \dots \quad (2) \end{aligned}$$

Substituting this value of x from (3) in (1), we get

$$y = \frac{a}{2} \left[\ln \frac{(p-1)}{\sqrt{(1+p^2)}} - \tan^{-1} p \right] + C + \operatorname{atan}^{-1} p$$

$$\text{or, } y = \frac{a}{2} \left[\ln \frac{(p-1)}{\sqrt{(1+p^2)}} - \tan^{-1} p \right] + a \tan^{-1} p + C \dots (3)$$

Hence, equation (2) and (3) constitute the required solution in parametric form with p as parameter.

Method IV [Equation Homogeneous in x and y]

If the equation

$$f(x, y, p) = 0$$

is homogeneous in x and y it can be written as

$$F\left(\frac{dp}{dx}, \frac{y}{x}\right) = 0.$$

Then, it can be solved for $\frac{dy}{dx}$ using the method used for solving homogeneous differential equation of first order and first degree. On the other hand, if it is solvable for $\frac{y}{x}$, we can get

$$\begin{aligned} \frac{y}{x} &= g\left(\frac{dy}{dx}\right) \\ \Rightarrow y &= xg(p). \end{aligned}$$

Differentiation implies

$$\begin{aligned} p &= \frac{dy}{dx} = g(p) + xg'(p) \frac{dp}{dx} \\ \Rightarrow \frac{dx}{x} &= \frac{g'(p)dp}{p - g(p)}, \end{aligned}$$

which when integrated gives the required solution.

Example Solve $xy^2(p^2 + 2) = 2py^3 + x^2$

Solution

The given equation is

$$xy^2(p^2 + 2) = 2py^3 + x^2 \dots (1)$$

which is homogeneous in x and y .

This can be simplified as

$$\begin{aligned} xy^2p^2 + 2xy^2 &= 2py^3 + x^2 \\ \text{or, } xy^2p^2 - 2py^3 + 2xy^2 - x^2 &= 0 \end{aligned}$$

which is quadratic in p . So,

$$\begin{aligned} p &= \frac{2y^3 \pm \sqrt{4y^6 - 4 \cdot xy^2(2xy^2 - x^2)}}{2xy^2} \\ \text{or, } p &= \frac{2y^3 \pm \sqrt{4y^6 - 8x^2y^4 + 4x^4y^2}}{2xy^2} \\ \text{or, } p &= \frac{2y^3 \pm 2y\sqrt{y^4 - 2x^2y^2 + x^4}}{2xy^2} \\ \text{or, } p &= \frac{2y(y^2 \pm \sqrt{(y^2 - x^2)^2})}{2xy^2} \\ \text{or, } p &= \frac{(y^2 \pm (y^2 - x^2))}{xy} \end{aligned}$$

Taking +ve sign, we get

$$p = \frac{2y^2 - x^2}{xy} \dots (2)$$

And taking -ve sign, we get

$$p = \frac{x^2}{xy}$$

$$\text{or, } p = \frac{x}{y} \quad \dots \quad (3)$$

From (2)

$$p = \frac{2y^2 - x^2}{xy}$$

$$\frac{dy}{dx} = \frac{2y^2 - x^2}{xy}$$

which is homogeneous equation. So, we put

$$y = vx$$

$$\therefore \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Now, (2) Reduce to

$$v + x \frac{dv}{dx} = \frac{2v^2x^2 - x^2}{vx^2}$$

$$\text{or, } v + x \frac{dv}{dx} = \frac{2v^2 - 1}{v}$$

$$\text{or, } x \frac{dv}{dx} = \frac{2v^2 - 1}{v} - v$$

$$\text{or, } x \frac{dv}{dx} = \frac{2v^2 - 1 - v^2}{v}$$

$$\text{or, } x \frac{dv}{dx} = \frac{v^2 - 1}{v}$$

$$\text{or, } \frac{dx}{x} = \frac{v}{v^2 - 1} dv$$

Integration implies

$$\text{or, } \int \frac{dx}{x} = \int \frac{v}{v^2 - 1} dv$$

$$\text{or, } \ln x = \frac{1}{2} \int \frac{2v}{v^2 - 1} dv$$

$$\text{or, } \ln x = \frac{1}{2} \ln(v^2 - 1) + \ln C$$

$$\text{or, } \ln x = \ln \sqrt{v^2 - 1} + \ln C$$

$$\text{or, } \ln x = \ln C \sqrt{v^2 - 1}$$

$$\text{or, } x = C \sqrt{v^2 - 1}$$

$$\text{or, } x = C \sqrt{\left(\frac{y}{x}\right)^2 - 1}$$

$$\text{or, } x = C \frac{\sqrt{y^2 - x^2}}{x}$$

$$\text{or, } x^2 = C \sqrt{y^2 - x^2}$$

$$\text{or, } x^4 = C^2(y^2 - x^2)$$

$$\therefore x^4 - C^2(y^2 - x^2) = 0 \quad \dots \quad (A)$$

From (3), we have

$$\begin{aligned} \text{or, } p &= \frac{x}{y} \\ \text{or, } \frac{dy}{dx} &= \frac{x}{y} \\ \text{or, } x dx &= y dy \end{aligned}$$

Integration implies

$$\begin{aligned} \int x dx &= \int y dy \\ \text{or, } \frac{x^2}{2} &= \frac{y^2}{2} + \frac{C}{2} \\ \text{or, } x^2 - y^2 - C &= 0 \quad \dots \quad (B) \end{aligned}$$

Combining (A) and (B), we get the combined general solution as

$$[x^4 - C^2(y^2 - x^2)][x^2 - y^2 - C] = 0$$

where, C is some arbitrary constant.

Method V [Clairaut's Equations]

Definition

An equation of the form

$$y = px + f(p)$$

is known as Clairaut's equation.

General Solution of Clairaut's Equation

Show that the general solution of Clairaut's equation

$$y = px + f(p)$$

is

$$y = cx + f(c)$$

where, c is an arbitrary constant.

Proof

The Clairaut's equation is

$$y = px + f(p) \quad \dots \quad (1)$$

Differentiating (1) w.r.t. x , we get

$$\begin{aligned} \frac{dy}{dx} &= p + x \frac{dp}{dx} + f'(p) \frac{dp}{dx} \\ \text{or, } p &= p + [x + f'(p)] \frac{dp}{dx} \\ \text{or, } [x + f'(p)] \frac{dp}{dx} &= 0 \end{aligned}$$

Ignoring the factor $x + f'(p)$, which does not involve $\frac{dp}{dx}$, we get

$$\begin{aligned} \frac{dp}{dx} &= 0 \\ \rightarrow dp &= 0 \end{aligned}$$

Integration implies

$$p = c,$$

c being an arbitrary constant.

Putting this value of p in (1), the required solution of (1) is

$$y = cx + f(c).$$

Note

If we eliminate p between

$$x + f'(p) = 0 \text{ and (1),}$$

we shall arrive at a solution,

$$y = cx + f(c),$$

free from an arbitrary constant, which is not a particular case of the general solution. Such a solution is called a singular solution of (1),

Example

1. Solve (Find the complete and singular solutions) the equation $y = px + p - p^2$.

Solution

The given equation is

$$y = px + p - p^2 \quad \dots \quad (1)$$

This is Clairaut's equation.

Differentiating (1) w.r.t. x , we get

$$\begin{aligned} \frac{dy}{dx} &= p + x \frac{dp}{dx} + \frac{dp}{dx} - 2p \frac{dp}{dx} \\ \text{or, } p &= p + (x + 1 - 2p) \frac{dp}{dx} \\ \text{or, } (x + 1 - 2p) \frac{dp}{dx} &= 0 \end{aligned}$$

Either

$$\frac{dp}{dx} = 0 \quad \dots \quad (2)$$

Or,

$$x + 1 - 2p = 0 \quad \dots \quad (3)$$

From equation (2), we get

$$\begin{aligned} \frac{dp}{dx} &= 0 \\ \rightarrow dp &= 0 \end{aligned}$$

Integration implies

$$p = c.$$

Substituting $p = c$ in (1), we get the complete (general) solution of (1) as

$$y = cx + c - c^2$$

Again, (3) implies

$$\begin{aligned} 2p &= x + 1 \\ p &= \frac{x + 1}{2} \end{aligned}$$

Eliminating p from (1) and (3), we get

$$\begin{aligned} y &= x \left(\frac{x + 1}{2} \right) + \frac{x + 1}{2} - \left(\frac{x + 1}{2} \right)^2 \\ \text{or, } y &= \left(\frac{x + 1}{2} \right) \left[x + 1 - \frac{x + 1}{2} \right] \\ \text{or, } y &= \left(\frac{x + 1}{2} \right) \left(\frac{x + 1}{2} \right) \\ \text{or, } y &= \left(\frac{x + 1}{2} \right)^2 \\ \text{or, } y &= \frac{(x + 1)^2}{4} \\ \text{or, } 4y &= (x + 1)^2 \end{aligned}$$

Hence,

$$4y = (x + 1)^2$$

is the required singular solution.

2. Find the general and singular solution of

$$(y - px)^2 (1 + p^2) = a^2 p^2$$

Solution

The given equation is

$$(y - px)^2 (1 + p^2) = a^2 p^2 \quad \dots \quad (1)$$

$$\text{or, } (y - px)^2 = \frac{a^2 p^2}{1 + p^2}$$

$$\text{or, } y - px = \pm \frac{ap}{\sqrt{1 + p^2}}$$

$$\text{or, } y = px \pm \frac{ap}{\sqrt{1 + p^2}} \quad \dots \quad (2)$$

which is Clairaut's equation. Differentiating w.r.t. x , we get

$$\frac{dy}{dx} = p + x \frac{dp}{dx} + \left[\frac{\sqrt{1 + p^2} \frac{dp}{dx} \cdot a - ap \frac{1}{2\sqrt{1 + p^2}} 2p \cdot \frac{dp}{dx}}{1 + p^2} \right]$$

$$p = p + x \frac{dp}{dx} + \left[\frac{\sqrt{1 + p^2} \frac{dp}{dx} \cdot a - \frac{ap^2}{\sqrt{1 + p^2}} \cdot \frac{dp}{dx}}{1 + p^2} \right]$$

$$x \frac{dp}{dx} + a \left[\frac{(1 + p^2 - p^2) \cdot \frac{dp}{dx}}{(1 + p^2)^{\frac{3}{2}}} \right] = 0$$

$$\left[x + \frac{a}{(1 + p^2)^{\frac{3}{2}}} \right] \frac{dp}{dx} = 0$$

Therefore, either

$$\frac{dp}{dx} = 0 \quad \dots \quad (2)$$

Or,

$$x + \frac{a}{(1 + p^2)^{\frac{3}{2}}} = 0 \quad \dots \quad (3)$$

From (2)

$$\begin{aligned} \frac{dp}{dx} &= 0 \\ \rightarrow dp &= 0 \end{aligned}$$

Integration implies

$$p = c.$$

Substituting $p = c$ in (1), we get

$$(y - cx)^2 (1 + c^2) = a^2 c^2,$$

which is the required general solution of (1).

From (3)

$$x + \frac{a}{(1+p^2)^{\frac{3}{2}}} = 0$$

$$\text{or, } x = -\frac{a}{(1+p^2)^{\frac{3}{2}}} \quad \dots \quad (4)$$

$$\text{or, } x^{\frac{2}{3}} = \frac{a^{\frac{2}{3}}}{1+p^2} \quad \dots \quad (A)$$

Using (4) in (2), we get

$$y = -\frac{ap}{(1+p^2)^{\frac{3}{2}}} + \frac{ap}{\sqrt{1+p^2}}$$

$$\text{or, } y = ap \left[\frac{-1 + 1 + p^2}{(1+p^2)^{\frac{3}{2}}} \right]$$

$$\text{or, } y = \frac{ap^3}{(1+p^2)^{\frac{3}{2}}}$$

$$\text{or, } y^{\frac{2}{3}} = \frac{a^{\frac{2}{3}}p^2}{1+p^2} \quad \dots \quad (B)$$

Adding (A) and (B) to eliminate p , we get

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = \frac{a^{\frac{2}{3}}}{1+p^2} + \frac{a^{\frac{2}{3}}p^2}{1+p^2} = \frac{a^{\frac{2}{3}}(1+p^2)}{1+p^2} = a^{\frac{2}{3}}$$

Therefore, the required singular solution of (1) is

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}.$$

Method VI (Equations Reducible to Clairaut's Form)

Some equation can be reduced to Clairaut's form by suitable substitutions. We discuss on the following three different forms:

Form I

Equations of the form

$$y^2 = \left(p \frac{y}{x}\right)x^2 + f\left(p \frac{y}{x}\right) \quad \dots \quad (1)$$

For solving such an equation, we put

$$x^2 = u \text{ and } y^2 = v$$

Now, differentiation implies

$$2xdx = du \text{ and } 2ydy = dv$$

Therefore, by division, we get

$$\frac{2ydy}{2xdx} = \frac{dv}{du}$$

$$\text{or, } \frac{ydy}{xdx} = \frac{dv}{du}$$

$$\text{or, } \frac{py}{x} = q, \text{ where } q = \frac{dv}{du}$$

Hence, the given equation becomes

$$v = qu + f(q) \quad \dots \quad (2)$$

which is in Clairaut's form. So, its solution is obtained as before,

$$\begin{aligned} i.e. v &= cu + f(c) \\ \Rightarrow y^2 &= cx^2 + f(c) \quad \blacksquare \end{aligned}$$

Example

Solve $x^2(y - px) = yp^2$

Solution

The given equation is

$$x^2(y - px) = yp^2 \quad \dots \quad (1)$$

This equation can be written as

$$\begin{aligned} yp^2 + x^3p - x^2y &= 0 \\ or, \quad x^2y - x^3p &= yp^2 \\ or, \quad x^2(y - px) &= yp^2 \\ or, \quad (y - px) &= \frac{(yp^2)}{x^2} \\ or, \quad y^2 - pxy &= \frac{y^2p^2}{x^2} \\ or, \quad y^2 &= pxy + \left(\frac{py}{x}\right)^2 \\ or, \quad y^2 &= \left(p\frac{y}{x}\right)x^2 + \left(\frac{py}{x}\right)^2 \quad \dots \quad (2) \end{aligned}$$

Now,

put $x^2 = u$, any $y^2 = v$

So, differentiation implies

$$2xdx = du \text{ and } 2ydy = dv$$

By division, we get

$$\begin{aligned} \frac{2ydy}{2xdx} &= \frac{dv}{du} \\ \rightarrow \frac{y}{x}p &= \frac{dv}{du} \\ \rightarrow \frac{py}{x} &= q, \quad \text{where } q = \frac{dv}{du} \end{aligned}$$

Substituting these values, equation (2) reduces to

$$v = qu + q^2 \quad \dots \quad (3)$$

which is in Clairaut's form. So, using previous method, the solution of (3) is

$$v = cu + c^2,$$

where c is some arbitrary constant.

$$or, \quad y^2 = cx^2 + c^2$$

Hence, the required solution of (1) is

$$y^2 = cx^2 + c^2 \quad \blacksquare$$

Form II

Equation of the Form $e^{by}(a - bp) = f(pe^{by-ax})$

To solve an equation of this form, we put

$$e^{ax} = u \text{ and } e^{by} = v.$$

Example

Solve $e^{4x}(p - 1) + e^{2y}p^2 = 0$

Solution

The given equation is

$$e^{4x}(p-1) + e^{2y}p^2 = 0 \quad \dots \quad (1)$$

$$\rightarrow p-1 = p^2 e^{2y-4x}$$

$$\rightarrow e^{2y}(p-1) = (pe^{2y-2x})^2 \quad \dots \quad (2)$$

which is of the form II, where $a = 2, b = 2$.

Now,

put $e^{2x} = u$ and $e^{2y} = v$.

Therefore,

$$2e^{2x}dx = du \text{ and } 2e^{2y}dy = dv$$

and by division we get

$$\frac{2e^{2y}dy}{2e^{2x}dx} = \frac{dv}{du}$$

$$\text{or, } \frac{v}{u}p = q, \quad \text{where } q = \frac{dv}{du}$$

$$\text{or, } p = \frac{uq}{v}$$

Substituting these values in (1), we get

$$e^{4x}(p-1) + e^{2y}p^2 = 0$$

$$\rightarrow u^2 \left(\frac{uq}{v} - 1 \right) + v \left(\frac{u^2 q^2}{v^2} \right) = 0$$

$$\rightarrow uq - v + q^2 = 0$$

$$\rightarrow v = uq + q^2 \quad \dots \quad (3)$$

which is in Clairaut's form. Solving by the previous method, the solution of (3) is

$$v = uc + c^2$$

$$\text{or, } e^{2y} = ce^{2x} + c^2,$$

where, c is an arbitrary constant. Hence, the required solution of (1) is

$$e^{2y} = ce^{2x} + c^2 \quad \blacksquare$$

Form III

[Substitution of $y^2 = v$]

Example

Solve $y = 2px - ayp^2$.

Solution

The given equation is

$$y = 2px - ayp^2 \quad \dots \quad (1)$$

Multiplying both sides by y , (1) reduces to

$$y^2 = 2pxy - ay^2p^2$$

$$y^2 = (2py)x - \frac{a}{4}(2yp)^2 \quad \dots \quad (2)$$

Put $y^2 = v$.

$$\therefore 2y \frac{dy}{dx} = \frac{dv}{dx}$$

$$2yp = q, \quad \text{where } q = \frac{dv}{dx}$$

Then, (2) reduces to

$$v = qx - \frac{a}{4}q^2$$

which is in Clairaut's form.

So, by previous method, its solution is

$$v = cx - \frac{a}{4}c^2$$

$$\text{or, } y^2 = cx - \frac{ac^2}{4},$$

where c is an arbitrary constant. Hence, the required solution of (1) is

$$y^2 = cx - \frac{ac^2}{4} \quad \blacksquare$$